
Exploring Informational Content and Semantic Structure in Political Manifestos: Evidence from the 1988 French Legislative Elections

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Abstract

Political discourse is often characterized by abstract, generic, and formulaic expressions, commonly referred to as “wooden language” or “langue de bois”. This paper proposes an exploratory Natural Language Processing approach to quantify this phenomenon in political manifestos from the Archelec corpus, focusing on the 1988 French legislative elections. The analysis combines corpus exploration, semantic density measurement, lexical similarity through TF-IDF, embedding-based semantic similarity, document ranking, and clustering. Results show that semantic density provides a useful proxy for distinguishing more informative texts from more abstract ones, while embedding-based representations capture deeper relationships between manifestos than TF-IDF. Clustering analysis further suggests that political manifestos are not randomly distributed in semantic space, but form coherent groups that may reflect thematic or rhetorical structures.

1 Introduction

Political manifestos are an important source for studying electoral communication. They provide information about candidates, political priorities, rhetorical strategies, and ideological positioning. However, political texts do not only differ in terms of explicit policy content. They also vary in their level of concreteness, informational density, and abstraction.

A recurring feature of political communication is the use of vague or formulaic language. This phenomenon is often described as “wooden language” or, in French, “langue de bois”. Such language can be characterized by abstract expressions, repetitive formulas, and generic claims that appear meaningful but may contain limited concrete information.

The objective of this project is to investigate whether this phenomenon can be explored quantitatively using Natural Language Processing techniques. Rather than manually classifying political manifestos, the project proposes a computational approach based on semantic density and semantic similarity.

The analysis focuses on four main questions. First, how heterogeneous is the corpus in terms of document length and vocabulary? Second, can semantic density provide a useful approximation of informational content? Third, how do lexical and semantic similarity methods differ when applied to political manifestos? Fourth, can clustering reveal latent structure in the corpus?

To address these questions, the project combines exploratory data analysis, semantic density computation, TF-IDF representations, sentence embeddings, cosine similarity, document ranking, PCA visualization, and K-Means clustering.

2 Background and Related Work

Traditional approaches to text representation often rely on frequency-based methods. One of the most widely used techniques is TF-IDF, which represents documents according to the importance of words

within a corpus. TF-IDF is useful because it is interpretable and computationally efficient. However, it is limited because it mainly captures lexical overlap.

In political text analysis, this limitation is important. Two documents may express similar political ideas using different vocabulary. Conversely, two texts may share many words while expressing different meanings. For this reason, lexical similarity alone may not be sufficient to capture deeper relationships between political manifestos.

Recent advances in NLP rely on dense vector representations generated by transformer-based models. Sentence embeddings represent texts in a semantic space where documents with similar meanings tend to be close to one another. These models can capture contextual information and are therefore more suitable for semantic similarity tasks.

In this project, TF-IDF is used as a baseline, while sentence embeddings are used to capture semantic proximity between documents. The comparison between the two approaches helps evaluate whether embeddings provide richer information than lexical methods in the analysis of political discourse.

3 Dataset and Corpus Description

The analysis is based on the Archelec corpus, which contains political manifestos from the 1988 French legislative elections. The corpus includes several thousand documents produced by candidates from different political parties and regions, providing a broad representation of electoral discourse at the national level.

These manifestos correspond to official campaign materials and therefore reflect how candidates choose to present their priorities, positions, and rhetorical strategies to voters. As such, the dataset offers a relevant entry point for studying political language in a real-world context.

The texts were digitized using Optical Character Recognition (OCR). As a result, the dataset contains noise such as inconsistent spacing, formatting errors, missing characters, and spelling inconsistencies. While preprocessing reduces some of these issues, it does not fully eliminate them.

Focusing on a single election year ensures temporal consistency, but also limits the scope of the analysis to a specific political context.

3.1 Corpus Overview

The dataset contains 3628 documents, each corresponding to one manifesto. Document length varies significantly, which motivates an initial exploratory analysis.

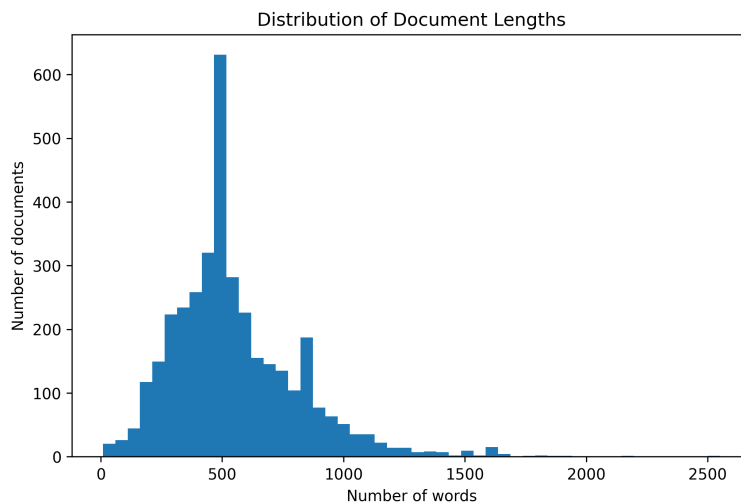


Figure 1: Distribution of document lengths in the corpus.

The distribution shows a clear concentration of documents in a moderate length range, alongside a smaller number of significantly longer texts. This heterogeneity is important, as document length can influence both lexical and semantic representations.

Longer texts tend to include more diverse vocabulary and potentially multiple themes, while shorter texts are often more focused but may contain less information overall.

3.2 Frequent Vocabulary

As a preliminary lexical analysis, the most frequent words in the cleaned corpus are examined.

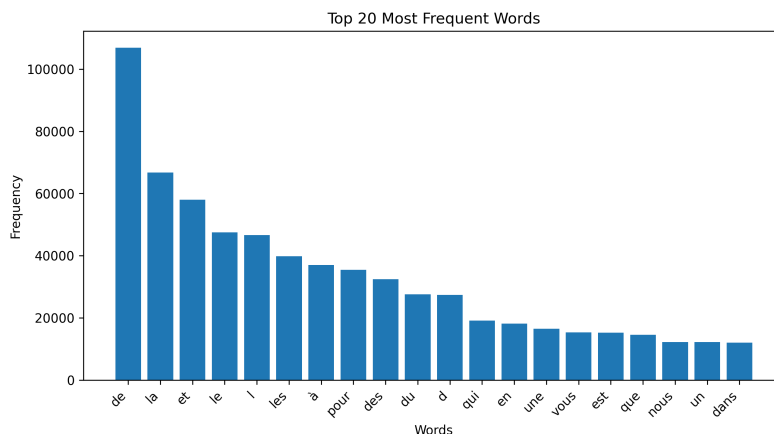


Figure 2: Most frequent words in the cleaned corpus.

The frequency distribution highlights the repetitive nature of political language. Many frequent terms correspond to institutional, electoral, or rhetorical vocabulary.

This suggests that political discourse relies on a relatively stable set of expressions, reflecting a degree of standardization in how candidates communicate.

However, word frequency alone does not capture meaning. This motivates the use of semantic representations in the following sections.

4 Methodology

4.1 Preprocessing

Before applying any NLP methods, the text is cleaned in a relatively simple way. This includes lowercasing, removing non-alphabetic characters, and normalizing whitespace.

Given that the dataset comes from OCR, the objective is not to obtain perfectly clean text, but rather to reduce the most obvious sources of noise. OCR artifacts such as irregular spacing, broken words, or corrupted characters can affect both frequency-based methods and embedding representations.

The preprocessing step therefore aims at finding a balance: reducing noise while preserving as much of the original content as possible. This is particularly important in political texts, where stylistic variation may itself carry information.

4.2 Semantic Density

Semantic density is introduced as a simple proxy for how informative a text is. The underlying idea is that texts containing a higher proportion of content-bearing words tend to be more specific, while texts dominated by generic or functional language are often more abstract.

The metric is defined as:

$$\text{Semantic Density} = \frac{\text{Number of content words}}{\text{Total number of words}} \quad (1)$$

In practice, content words are approximated using lexical filtering rather than full linguistic annotation. While this approach is less precise than using part-of-speech tagging, it remains computationally efficient and sufficient for a first exploration of the data.

Semantic density should not be interpreted as an absolute measure. Instead, it is used in a comparative sense, to highlight differences between documents and to identify variations in informational content across the corpus.

4.3 TF-IDF Baseline

TF-IDF is used as a baseline representation. It reflects how important a word is within a document relative to the whole corpus, giving more weight to words that are frequent in a given document but rare overall.

This approach is particularly effective for identifying documents that share specific vocabulary. However, it does not account for semantic equivalence. Two texts that express similar ideas using different words may appear unrelated in a TF-IDF representation.

For this reason, TF-IDF is mainly used as a reference point, allowing us to compare lexical similarity with more advanced semantic approaches.

4.4 Sentence Embeddings

To move beyond lexical similarity, each document is represented using sentence embeddings.

These embeddings are generated using a pretrained Sentence Transformer model, which maps each manifesto to a dense vector in a high-dimensional semantic space. In this space, documents that are similar in meaning are expected to be located closer to each other.

Unlike TF-IDF, embeddings are able to capture contextual relationships between words and sentences. This makes them particularly suitable for analyzing political texts, where similar ideas may be expressed through different formulations.

4.5 Document Ranking

Similarity is further explored through a ranking task. Given a query document, the most similar documents are retrieved using cosine similarity.

This is done both for TF-IDF and embedding-based representations. The comparison is useful because it highlights how the notion of similarity changes depending on the underlying representation.

In addition to numerical similarity scores, the ranking results are inspected qualitatively. This helps assess whether retrieved documents are actually coherent in terms of content, rather than simply similar in terms of vocabulary.

4.6 Clustering

To explore the global structure of the corpus, K-Means clustering is applied to the embedding space.

Clustering is used in an exploratory way, with the goal of identifying whether documents naturally form groups. It is not intended to produce definitive classifications, but rather to provide insight into how texts are distributed in semantic space.

To make these patterns easier to interpret, Principal Component Analysis (PCA) is used to project the high-dimensional embeddings into two dimensions. Although this projection involves a loss of information, it provides a useful visualization of the overall structure of the data.

The combination of clustering and visualization allows for a more intuitive understanding of how documents relate to each other beyond pairwise similarity.

5 Experiments

The analysis is organized across four complementary notebooks, each addressing a different aspect of the problem while contributing to a coherent overall exploration of the corpus.

The starting point is a general exploration of the dataset. Basic statistics such as document length and vocabulary distribution are examined in order to identify heterogeneity in the corpus and to anticipate how this variability may affect later steps of the analysis.

The introduction of semantic density follows naturally from this initial exploration. Rather than aiming at a precise linguistic measure, the objective is to test whether a simple proxy can already capture differences in how informative or abstract the texts are. The focus is therefore on the distribution of the metric and on how it varies across documents.

The analysis then moves to similarity. Two different representations are compared: TF-IDF, which captures lexical overlap, and embedding-based representations, which operate at the semantic level. Instead of relying only on numerical outputs, the comparison also includes qualitative inspection through ranking results, in order to better understand how the notion of similarity changes depending on the representation.

Finally, the embedding space is further explored using clustering and visualization techniques. PCA is used to obtain a low-dimensional projection that makes the structure of the data easier to interpret, while K-Means clustering provides a way to identify potential groupings within the corpus.

Because embedding computation and pairwise similarity are computationally expensive, part of the analysis is performed on a random subset of documents. This represents a practical compromise: while the full dataset would be preferable, the subset remains large enough to preserve the main patterns of the corpus while allowing for faster experimentation.

The different outputs produced in these experiments — including distributions, similarity matrices, ranking results, and projections — should be seen as complementary rather than independent. Together, they provide multiple perspectives on the same underlying structure, making it easier to interpret the results and relate them back to the initial questions.

6 Results

6.1 Semantic Density

The distribution of semantic density suggests that most documents are concentrated within a relatively narrow range, but still exhibit meaningful variation across the corpus.

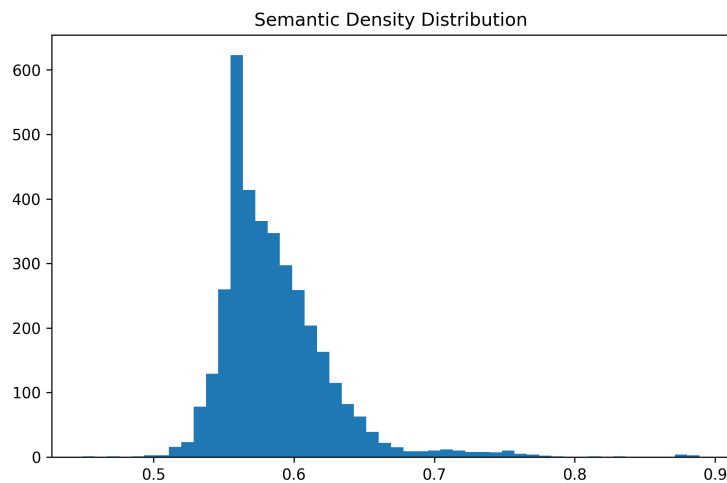


Figure 3: Distribution of semantic density across documents.

At first glance, the distribution appears relatively compact, which indicates that a large proportion of manifestos share similar structural characteristics. However, a more careful inspection reveals that the distribution is not perfectly symmetric. In particular, the lower tail extends slightly further than the upper one, suggesting that a subset of documents is characterized by noticeably lower density values.

This asymmetry is interesting because it points to differences that are not immediately visible when simply reading individual documents. While many manifestos follow a common template, some rely more heavily on abstract or repetitive formulations.

Documents with higher semantic density tend to include a greater number of content-bearing elements. These often take the form of references to local issues, institutions, concrete proposals, or specific actors. As a result, these texts appear more grounded in context and provide a clearer sense of what is being discussed.

On the other hand, lower-density documents are typically dominated by more general expressions. Instead of introducing new or specific information, they often rely on broad statements and recurring rhetorical patterns. This type of language aligns closely with what is usually described as “wooden language”, where the emphasis is more on form than on content.

It is important to stress that semantic density is only a proxy. It does not capture deeper linguistic features such as argument structure, coherence, or persuasive strategies. Nevertheless, even in this simplified form, it provides a useful way to highlight differences across documents and to identify potential variations in informational content.

6.2 Similarity Analysis

The comparison between TF-IDF and embedding-based similarity highlights a clear difference in how relationships between documents are captured.

TF-IDF focuses on lexical overlap. This means that documents are considered similar when they share common words, especially those that are relatively rare in the corpus. This approach works well when similar ideas are expressed using similar vocabulary. However, it quickly becomes limiting when the same concepts are expressed using different wording.

Embedding-based similarity provides a more flexible alternative. By representing each document in a semantic space, it allows similarity to be measured in terms of meaning rather than exact word matching. This is particularly relevant in the context of political discourse, where similar messages can be conveyed through different formulations.

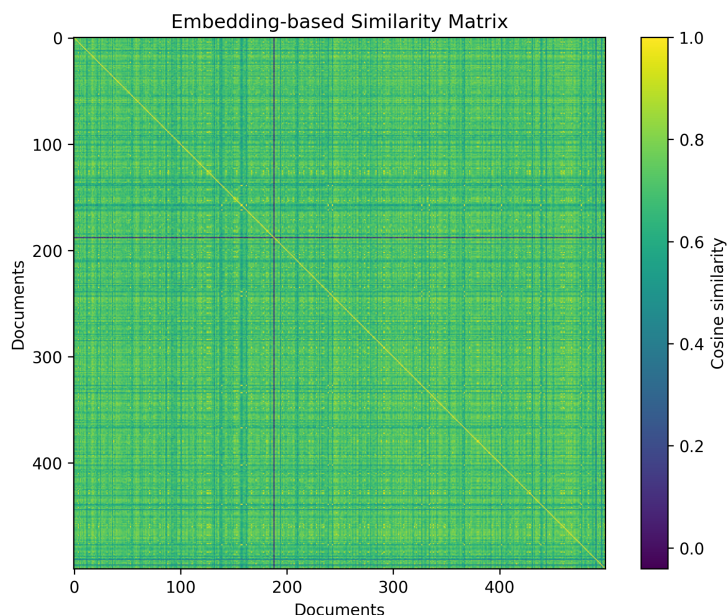


Figure 4: Embedding-based cosine similarity matrix.

The heatmap provides a global view of pairwise similarities. Rather than appearing random, the matrix exhibits visible structure. Some regions are consistently darker, indicating groups of documents that are more similar to each other.

This suggests that the corpus is not homogeneous. Instead, it contains subsets of documents that share underlying similarities, even when these are not immediately obvious at the lexical level.

6.3 Similarity Score Distribution

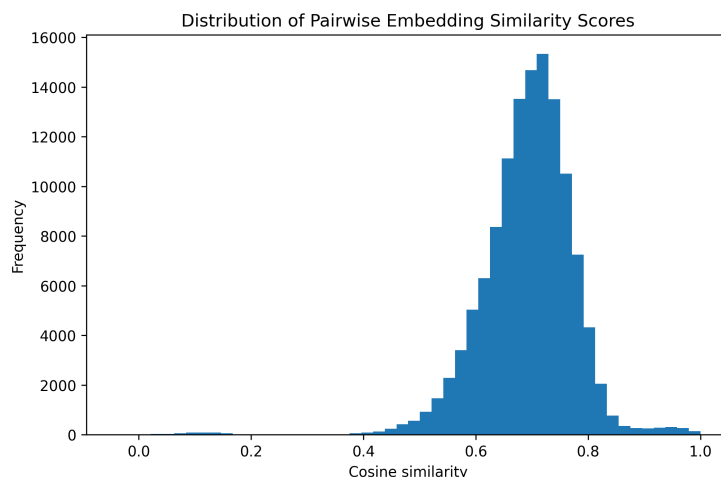


Figure 5: Distribution of pairwise embedding similarity scores.

The distribution of similarity scores provides additional insight into the structure of the corpus. Most document pairs fall within an intermediate similarity range, while only a limited number of pairs exhibit very high or very low similarity.

This pattern indicates that the corpus is neither highly uniform nor completely fragmented. Instead, documents are organized along a continuum of semantic proximity.

In practical terms, this means that while many manifestos share a baseline level of similarity, there are still noticeable differences that separate more closely related documents from those that are more distant.

The absence of extreme values for most pairs also suggests that political discourse tends to operate within a relatively constrained semantic space, where texts are neither identical nor completely unrelated.

6.4 Document Ranking

The ranking task offers a more concrete way to evaluate similarity methods.

When using TF-IDF, the retrieved documents tend to share specific keywords with the query. This can produce reasonable matches, but it can also lead to results that are only superficially related, especially when common political terms are involved.

Embedding-based ranking behaves differently. It tends to retrieve documents that are closer in terms of overall meaning, even when the vocabulary differs. This often leads to more coherent results when inspected qualitatively.

For instance, documents discussing similar themes or issues may be retrieved together, even if they use different expressions. This highlights the advantage of embedding-based representations when dealing with heterogeneous textual data.

Although no formal evaluation metric is used here, the qualitative comparison between the two approaches consistently favors embeddings in terms of interpretability and relevance.

6.5 PCA Visualization of Embeddings

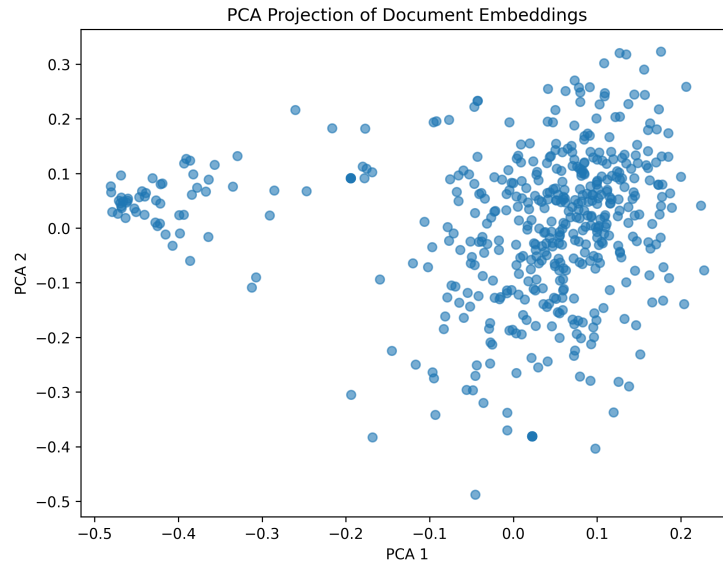


Figure 6: PCA projection of document embeddings.

The PCA projection offers a simplified view of the embedding space. Even though dimensionality reduction inevitably leads to a loss of information, it still reveals useful patterns.

Rather than being uniformly scattered, documents appear to form regions of higher density. This suggests that certain types of manifestos share similar representations, even without explicit grouping.

While PCA does not allow for precise interpretation of these regions, it confirms that the embedding space captures non-trivial variation across documents.

6.6 Clustering Analysis

Clustering provides a complementary perspective on this structure.

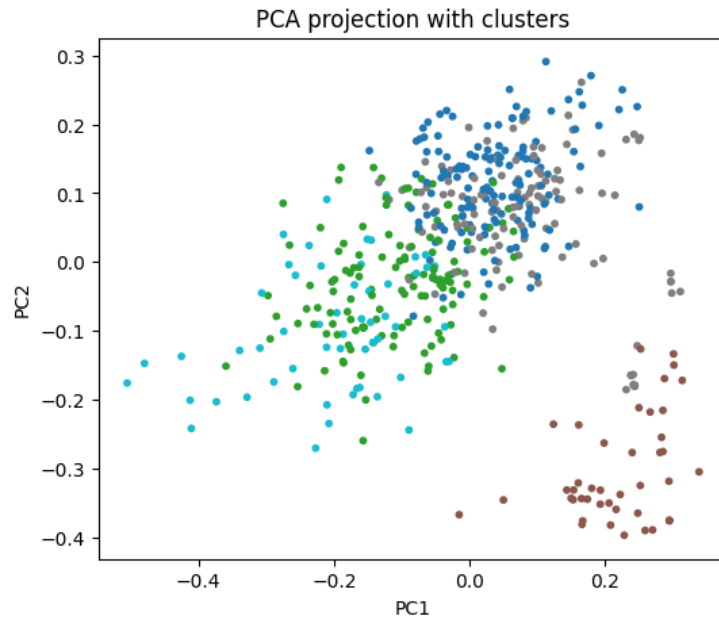


Figure 7: PCA projection of document embeddings colored by K-Means cluster.

The clusters are not perfectly separated, which is expected given the overlap in vocabulary, themes, and rhetorical strategies across political manifestos.

However, the clustering is not arbitrary either. The visualization shows that clusters tend to occupy distinct regions of the space, even if the boundaries remain diffuse.

This suggests that the model captures underlying differences between groups of documents, rather than simply partitioning the data randomly.

A closer inspection of cluster composition indicates that some groups are associated with more specific and locally grounded discourse, while others are dominated by more general or abstract language. This observation is consistent with the patterns already identified through semantic density.

Overall, clustering reinforces the idea that political manifestos exhibit structure that can be captured using embedding-based representations, even when this structure is not immediately visible.

7 Discussion and Interpretation

The results highlight the complementarity of the different methods used in the project, while also revealing some underlying patterns that deserve closer attention.

Semantic density, despite its simplicity, appears to capture a meaningful dimension of political discourse. Although the metric is not intended to be precise, it still reflects observable differences between documents. In particular, it helps distinguish between texts that are more specific and grounded in context, and those that rely more heavily on general or repetitive formulations.

This suggests that even a relatively basic measure can provide useful insight into variations in informational content across political manifestos. At the same time, the interpretation should remain cautious, as semantic density does not account for how information is structured, organized, or argued within a text.

A similar contrast emerges when comparing TF-IDF and embedding-based representations. While TF-IDF remains a valuable baseline—especially when documents share explicit vocabulary—it tends to emphasize surface-level similarities. Embeddings, in contrast, offer a more flexible notion of similarity, making it possible to identify relationships between texts that are not immediately visible at the lexical level.

This difference becomes particularly relevant in political discourse, where similar ideas can be expressed through different formulations. The qualitative inspection of ranking results further supports this point, as embedding-based retrieval generally produces outputs that are more coherent in terms of overall content.

Looking at the corpus more globally, the clustering results suggest that the data is not randomly organized. Instead, documents appear to follow underlying patterns that reflect differences in tone, specificity, or thematic focus. At the same time, the boundaries between clusters remain relatively diffuse. This is consistent with the nature of political texts, which tend to share common vocabulary and rhetorical conventions.

Rather than indicating a limitation of the method, this overlap can be interpreted as a reflection of the continuity of political discourse itself. Overall, these findings suggest that even relatively simple NLP techniques are sufficient to uncover non-trivial structure in political texts. While the analysis remains exploratory, it provides a useful basis for more systematic and refined investigations.

8 Limitations

The results should be interpreted with some caution, as several aspects of the analysis introduce potential sources of bias.

A first limitation comes from the nature of the dataset itself. The texts are obtained through OCR, which inevitably introduces noise. Errors in character recognition and formatting can affect tokenization and, more generally, the quality of the textual representations used throughout the analysis. Although basic preprocessing reduces some of these issues, it does not fully correct them.

Another important point concerns the use of semantic density. The metric is intentionally simple and does not aim to capture the full complexity of language. It ignores syntactic structure, argumentation, and discourse-level features, and therefore should not be interpreted as a direct measure of how “informative” or “good” a text is. Instead, it only provides a rough indication of differences across documents.

The use of pretrained embeddings also introduces limitations. While these models are generally robust, they are not specifically adapted to French political manifestos. As a result, certain domain-specific nuances or contextual meanings may not be fully captured in the embedding space.

In addition, the analysis focuses on a single election year. This choice ensures consistency, but it also limits the scope of the conclusions. The patterns observed here may depend on the specific political context of the 1988 elections and may not generalize to other periods or countries.

Finally, the clustering results remain exploratory. The number of clusters is fixed in advance, and their interpretation relies on qualitative inspection rather than external validation. Without additional information such as party affiliation or ideological positioning, it is difficult to assign precise meaning to each cluster.

For these reasons, the findings should be understood as indicative rather than definitive, and as a starting point for further investigation rather than a complete characterization of political discourse.

9 Conclusion

This project explores how relatively simple NLP techniques can be used to analyze political manifestos from both a quantitative and a qualitative perspective.

The results show that even basic measures, such as semantic density, are able to capture meaningful variation across documents. Although the metric remains approximate, it highlights differences that align with intuitive distinctions between more concrete and more abstract forms of political discourse.

The comparison between TF-IDF and embedding-based similarity further illustrates how lexical and semantic representations provide complementary insights. TF-IDF is effective in identifying documents that share vocabulary, while embeddings offer a more flexible view of similarity by capturing relationships at the level of meaning rather than wording.

Clustering and visualization techniques add another layer to the analysis. The fact that documents tend to occupy distinct regions in the embedding space suggests that political manifestos are not randomly distributed, but instead follow underlying patterns that may reflect differences in rhetorical style, thematic focus, or level of specificity.

These results also suggest that political discourse may be more structured than it appears at first glance. Even when different candidates use distinct vocabulary, they often converge toward similar semantic patterns, reflecting shared themes or strategic positioning within the political space.

From a political perspective, this indicates that variation across manifestos may be less about completely different ideas and more about differences in how similar messages are framed, ranging from concrete and localized proposals to more abstract and generalized rhetoric.

Although the analysis does not include explicit labels such as party affiliation, these qualitative patterns point toward the possibility of linking semantic structure to political positioning, provided that additional metadata is available.

More broadly, the project shows that computational approaches can complement traditional qualitative analysis. Rather than replacing interpretation, these methods provide a way to systematically explore large collections of political texts and to identify patterns that may not be immediately visible.

Future work could extend this analysis in several directions. Incorporating external information, such as party labels or electoral results, would make it possible to interpret clusters more precisely. Comparing multiple election years could also help track how political language evolves over time. Finally, more advanced linguistic features or domain-specific models could refine the analysis and capture subtler aspects of political communication.

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